

An expert system for the dyeing recipes determination

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This article deals with the modelization of the reasoning of an expert in dyeing. The dyeing operation is one of the latest stage in the whole textile process which alters the intrinsic properties of the articles. At present, there are expert systems for dyeing industries. Most of them have been developed by dyes and chemical auxiliary suppliers, and these tools are not widely used in dyeing factories. This paper describes the dyeing operation, and the rules of the expert in dyeing. The objects that he reasons about, his reasoning, and the critical points that he considers are also presented.

Keywords: Textile, expert system, dyeing, color

1. Introduction

Textile wet processing is very complex and diverse. It is carried out in multiple stages, and obtaining the desired color is just one simple step in the whole textile process.

During the reasoning of an expert in dyeing, many criteria are taken into account, and most of these criteria are subjective and very difficult to represent because they are related to color appreciation, and hand of textile evaluation. Most of the time, only a few people in the factory are able to solve the problems so that the knowledge belongs to a little part of the staff.

For these reasons, expert systems should be a high improvement. First, they can take the place of the expert when absent, and second, they permit the knowledge of the factory to be kept.

2. The textile finishing process

Dyeing, printing and finishing are the latest stages, in the textile process, that alter the intrinsic properties of textile materials, before they are handled or reassembled into final products.

A schematic representation of the whole process is given on Fig. 1. It consists in:

- a preparation operation. It could be often split into a scouring and a desizing operation, which remove the natural and acquired impurities from fibres, and into a bleaching operation,
- a coloring operation (dyeing and/or printing) using dyes or pigments as coloring agents. The desired color can be achieved by combining dyestuffs,
- a finishing operation to give specific properties to the material according to the clients' requirements: water-, oil repellent, crease resistance, etc. This operation is performed after the dyeing/printing operation and should not alter the shade.

In each step, many parameters have to be considered. So, even for specialists it is time-consuming and it requires a lot of knowledge.

The process of specifying color usually involves sending out physical standards to the dyehouse, that will dye out several samples accordingly and submit them to approval. The approved sample is then used as a guideline for bulk production. Color being a subjective perception, objectivity is of a great

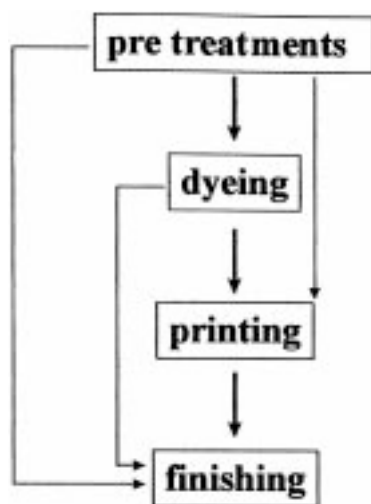


Fig. 1. The dyeing/finishing process.

importance. So, in the textile industry, the use of computer color matching is rather well established. The software are based on reflectance spectra, measured with a spectrophotometer, using specific equations. Numerical values can be calculated which permit the sample to be placed in a three-dimensional color order system known as “color space”. Such values enable color to be objectively compared using color difference formula (McDonald, 1997).

The calculation of a recipe of dyes is time consuming. So to restrict the number of possible recipes according to the requirements, a selection of dyes and processes has to be done by the dyer.

3. The existing expert systems

Many experts systems exist in the dyeing and finishing industry. Most of them have been developed by chemical products and dyestuffs suppliers (at the end of 80th and in the early of 90th). For example:

- Pré-matic (Günther, 1989) (CIBA): used for the bleaching of cotton fabrics.
- WOOLY (Frei, 1991) (SANDOZ): related to the determination of dyeing recipes for wool and polyamide/wool articles. The selection of dyes and dyeing processes is done by considering color fastness criteria, machines used, etc.

- BAFAREX (Lang, 1992) (BASF): developed for the determination of dyeing recipes for cotton and polyester/cotton articles, using vat and disperse dyes.
- OPTIMIST (Rüttiger, 1988) (BASF): used for the optimization of dyeing processes.
- TEXPERTO (Frei, 1992) (SANDOZ): this expert system fulfils finishing recipes.

Another expert system, also developed by SANDOZ, is used for the selection of fluorescent whiteners (Aspland, 1991).

These expert systems are often related to a specific field and only take the dyes and chemical products sold by these suppliers into account. So, it is an efficient tool for the technicians from these companies when they have to advice their clients. Unfortunately, they are not widely used in dyeing factories, because these factories often combined dyes and auxiliaries products from several suppliers.

So, an efficient expert system should be a tool, in which each factory could implement its own knowledge fitting the decision makers and the technical engineers.

4. The determination of a dyeing recipe

A “recipe of dyes” consists in one or many dyes, the quantity of each dyes (i.e., the quantity of product that should be used at the beginning of the dyeing) and the application process.

A “dyeing recipe” corresponds to a recipe of dyes, and indicates all the parameters of the dyeing process, such as the temperature, the volume of water, the quantity of each chemical auxiliary product, etc.

The determination of a dyeing recipe takes place in three main steps, as it is shown on Fig. 2.

The first step consists in selecting of the dyestuffs and of the dyeing process(es) that may be used to calculate the recipes of dyes. These selection is done by the dyer, by taking into account the client requirements.

The second step is the calculation of the recipes of dyes from the previously selected objects. This calculation is done from calibration range of dyes. A calibration range consists in dyeing an article, using a specific process and with several concentrations of dyes. Computers provide a mean of testing alternative dye combinations to find the optimum mixture that

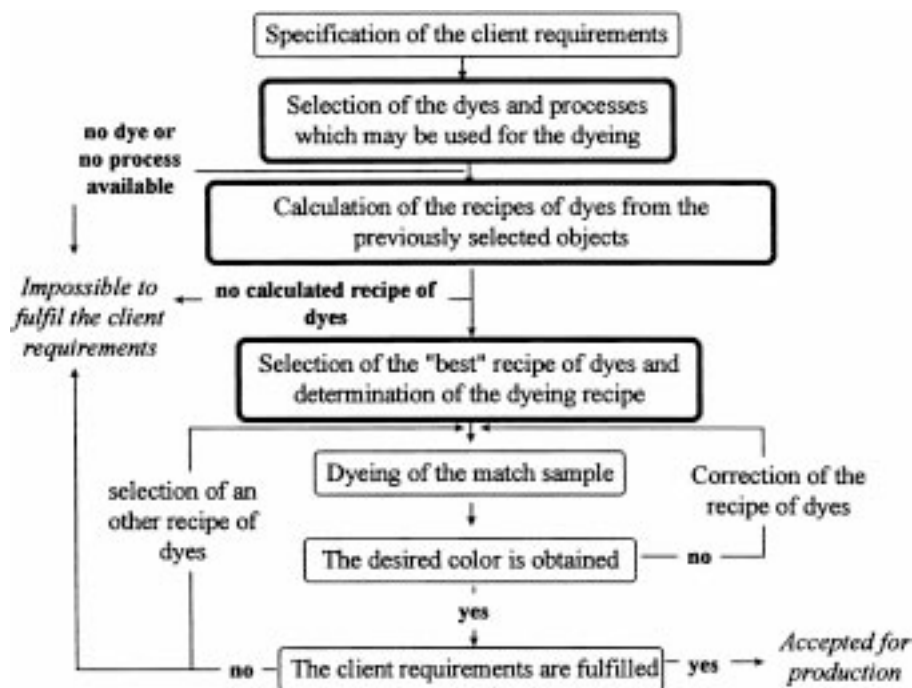


Fig. 2. Process of determination of a dyeing recipe.

will give minimum cost and/or color difference relative to the target. The number of recipes to be predicted rises very rapidly as the number of possible dye is increased. Then, it is very important to eliminate as soon as possible, all the objects that will not permit the requirements to be fulfilled. Moreover, in the case of articles made of several fibers, such as cotton/polyester articles, a recipe of dyes has to be calculated for cotton fibers, and an other one for the polyester fibres.

In the third step, the "best" calculated recipe of dyes is selected. The complete dyeing recipe is determined. Then a sample is dyed and if the dyed sample has the desired color and if the client requirements are fulfilled, this dyeing recipe is accepted for the production. If the desired color is not obtained, then the recipe of dyes has to be modified (i.e. the concentration of each dyes), until the color difference between the target and the match sample is negligible. Finally, if the care requirements are not achieved by the dyeing recipe (for instance, the washing fastness is too low), then the dyer must

choose an other recipe of dyes. Whenever there is some difficulties with all the calculated recipes of dyes, it means that the factory cannot fulfil the client specifications due to dyes or processes.

At present, there are tools that permit the recipes of dyes to be calculated (using the Kubelka-Munck's relationship) (McDonald, 1997). There are also some tools that are used for determining the dyeing recipes, considering the recipe of dyes (dyeing process and dyes used). This is done by rules defined for each process, such as:

If the concentration of dyes is lower than 5 g/l
Then the concentration of sodium chloride is equal to 20 g/l

So, the main difficulty is related to the first step. Usually some selection from the list of possible dye is made by the expert in dyeing before computation begins. Our work mainly consists in the modelization of this reasoning. At the end of this step, all the dyes that may not permit to fulfil the requirements should have been eliminated.

5. The clients requirements

The client's requirements may have several aspects. Nevertheless, it always consists in, at least:

- the article to be dyed (structure, composition, etc.),
- the desired color,
- the quantity of the article to be dyed.

Some other criteria may still be specified, such as:

- the color fastness requirements,
- the delivery time,
- the dimensional variations accepted, etc.

Most of these specifications depend on the end use of the textile materials. For example, if it is a furnishing item, then the dyed fabric must have a high fastness to light, whereas the fastness to washing is not so important.

6. The color fastness of a dyed fabric

This criterium is used to evaluate the behaviour of a dyed fabric to a specific test. The method currently used for assessments of fastness test results relies on comparing the contrast between tested and untested samples with those of the relevant one of internationally agreed physical scales, each consisting of pairs of grey chips having graded contrasts as it is shown on Fig. 3.

This is a visual evaluation, so it is very subjective, and for the same sample, it is possible to have a

difference between the evaluations done by several observers, equals to 1 or 2 points (McDonald, 1997). Moreover, most of the time, whatever the fastness test, the color fastness is not constant. It will vary with the concentration of dye on the fabric, and then with the concentration applied at the beginning of the dyeing.

7. The general process of reasoning of an expert in dyeing

The dyeing field is qualified of "opened" because the dyer must adapt his reasoning when new dyes, new processes are used. It is also qualified of "non-structured", because most of the time there is no evident way to obtain a solution.

The expert in dyeing often combined many kinds of knowledge to carry out the expertise. Then, the knowledge of a dyer is made of:

- Deep knowledge, which is the basic knowledge on the objects manipulated. It consists in relations between these objects and in rules issued from these relations. This is usually encyclopedical knowledge.
- Heuristic knowledge, gained from practical experiences, which consists in new relations between objects which do not have obvious direct links. It closely depends on the expert himself and on the factory.
- Meta-knowledge that governs the global process of reasoning, allowing to manipulate both deep and heuristic knowledge.

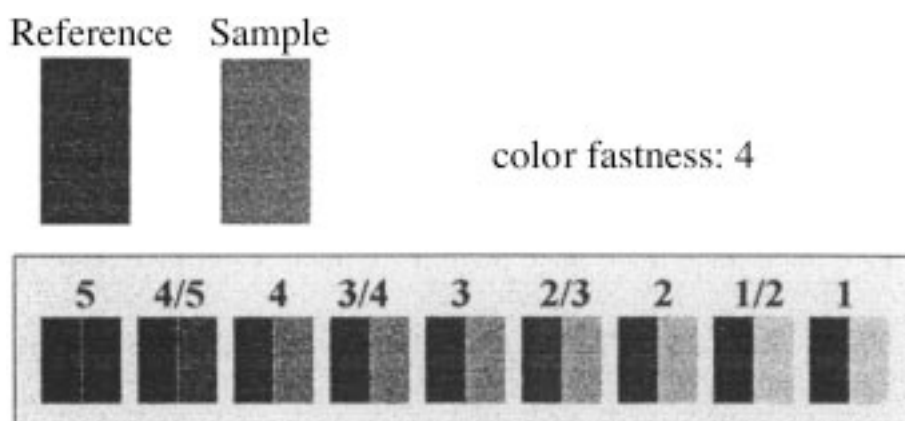


Fig. 3. Evaluation of the color fastness—case of the change in color related to a grey scale for the change in color.

8. The deep knowledge: the objects that are involved

In the finishing field, three main groups of objects are involved. These objects are represented on Fig. 4, using the semantic net method (a node represents a class of objects, and an arrow, the relation between two classes of objects).

These main groups of data are:

- the ARTICLES,
- the DYES,
- the PROCESSES.

8.1. The articles

A textile material is defined by several parameters:

- ⇒ the fibre(s) it is made of, and the proportion of each fibre,
- ⇒ its nature (yarn, woven-, non woven- or knitting fabric, ...),
- ⇒ its physical properties such as the breaking strength, etc.

Items sharing many characteristics are distributing into sub-groups of articles. Then, a sub-group of articles is defined as an collection of articles that are composed of the same fibres, and which have the same mechanical and physical properties, so that they can be dyed with the same machines. A group of articles is composed of sub-groups of articles made of the same fibres. Then, it is related to one or several fibres, whereas a sub-group of articles is related to only one group of articles, and an article is related to only one sub group of articles. Finally, one of the characteristic of a sub group of articles is the machine that can be used to dye them.

8.2. The dyes

The dyes are chemical compounds characterized by the chemical reactions that are involved when applied on a textile material (covalent bound, Van der Waals bound, etc.), and by the fibres they can color. That is why the dyestuffs are distributed into classes of dyes. Then, each dye is related to only one class of dyes. Nevertheless, even if dyes are related to the same class

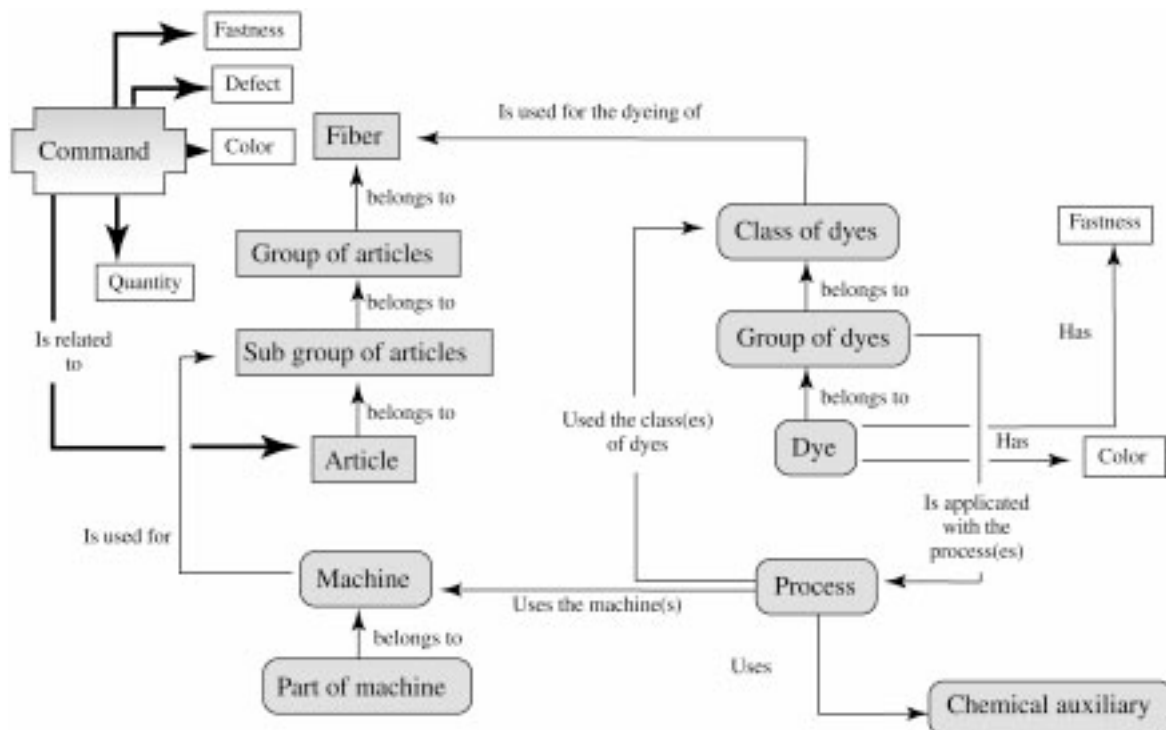


Fig. 4. Representation of the objects using the semantic net method—deep knowledge.

of dyes, they cannot necessarily be applied together, because they have not all the same chemical properties nor the same affinity for the fibres. These are also distributed into groups of dyes. Then, a group of dyes is made of dyes that are applied with the same processes.

So, a dye can be related to several group of dyes, whereas it is a part of only one class of dyes. The properties of dyes can be held with a range of dyes. A range of dyes consists in applying this dye, on the same article, with the same process and for several concentrations, as it is shown on Fig. 6. Then, for each concentration, the measurement of the color (in the form of reflectance spectra) will be used for the calculation of the recipe of dyes.

Moreover, each dye has fastness properties (to washing, to light, . . .). These properties depend on the dye, on the article, and especially of the fibers it is made and on the dyeing process.

8.3. The processes

A dyeing process is defined by:

- the successive machines that are used,
- the class(es) of dyes that is(are) applied,

- some parameters (temperature, volume of water, duration of the treatment, etc.) and auxiliary products that are attached to each part of machine.

In the example presented on Fig. 7, we considered a dyeing process for cotton/polyester articles. Two dyeing machines are successively used: the autoclave, for the dyeing of polyester fibres with disperse dyes, and Pad-Batch for the dyeing of cotton fibres with reactive dyes. The order of uses of these machines is crucial given that disperse dyes are applied in an acid medium and that some reactive dyes may be sensible to acid.

9. The deep knowledge: the causality connections between the objects

The reasoning of the expert in dyeing consists in eliminating all the objects that cannot fulfil clients requirements. The rules involved for this reasoning are issued from the causality connections between objects given by the semantic net (Fig. 4). Two kinds of rules are then used:

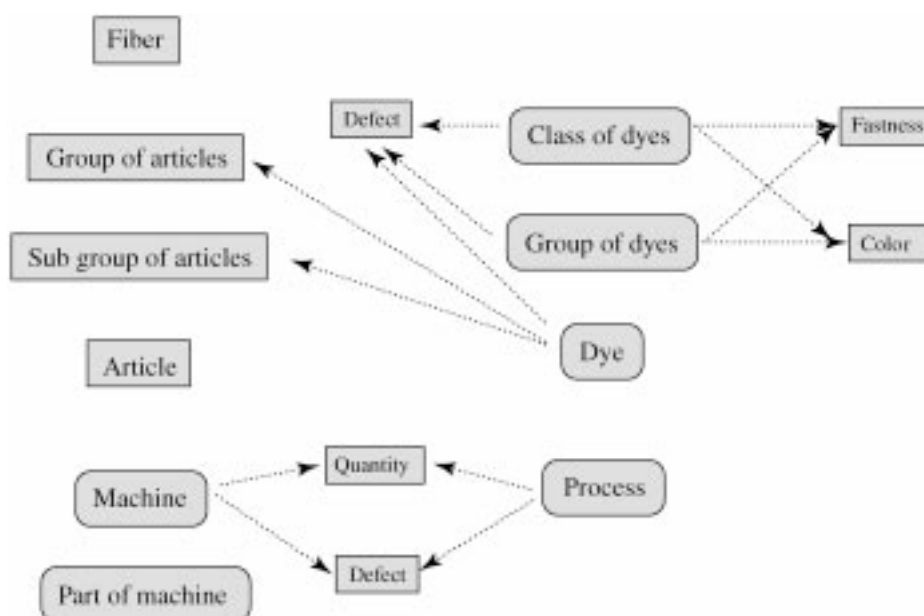


Fig. 5. Representation of the objects using the semantic net method — heuristic knowledge

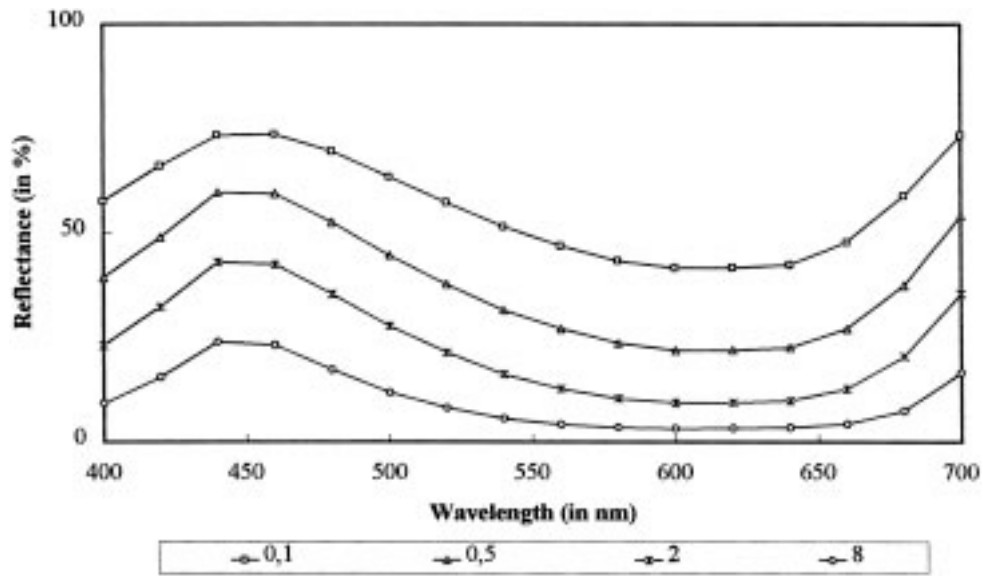


Fig. 6. A range of C.I. Reactive Blue 19 (concentration applied in %).

- “direct” elimination rules, that bring on the one hand a client requirement, and on the other hand elements of the semantic net (class of dyes, machine, etc.) into play. For example:

If A class of dyes cannot be used for at least one of the fibers that must be dyed
Then This class of dyes is unavailable for the case studied

- “indirect” elimination rules, that bring on the one hand an object from the semantic net, unavailable for the reasoning, and on the other hand elements from the semantic net, that are linked to this object (class of dyes, machine, etc) into play. All these rules are summarised in Table 1. For example:

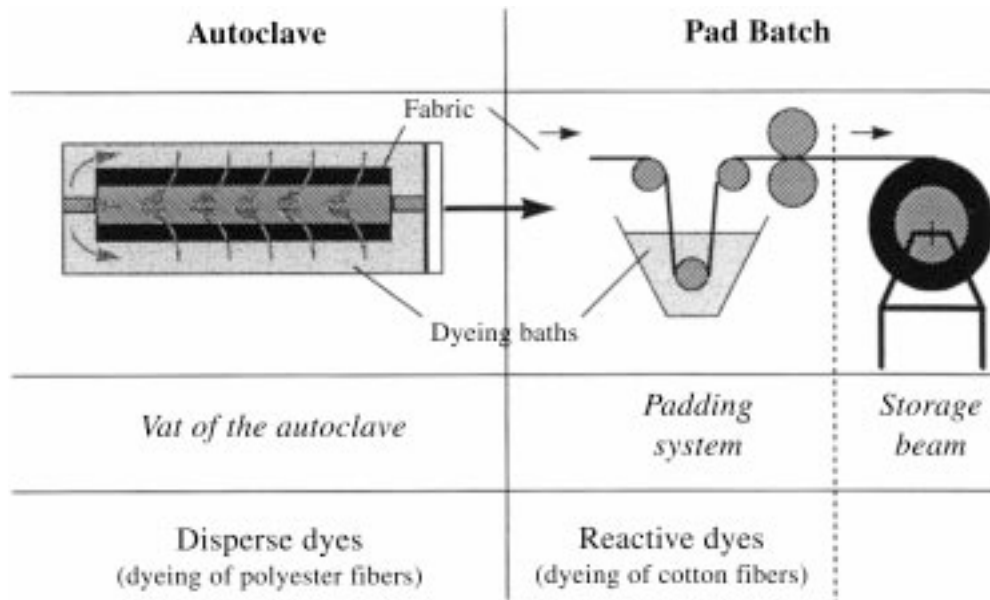


Fig. 7. A dyeing process—the example of the dyeing of polyester/cotton articles.

Table 1. The causality connections between the objects

<i>Object</i>	<i>Cause of elimination</i>
Class of dyes	This class of dyes cannot be used to dye at least one of the fibre that composed the article All the groups of dyes which are related to this class of dyes are unavailable All the dyeing processes that are used to applied this class of dyes are unavailable
Group of dyes	The class of dyes, that this group of dyes is related to, is unavailable All the dyes, in that group, are unavailable All the dyeing processes that could be used to applied that group of dyes have been eliminated
Dye	All the group of dyes, that this dye belong to, are unavailable At least one of the color fastness requirements cannot be fulfilled with this dye
Process	At least one of the class of dyes that the process can apply is unavailable All the group of dyes, that may be applied with this process, have been eliminated At least one of the machine that is used, in this dyeing process, is not available
Machine	This machine cannot be used for the treatment of the sub-group of articles, All the dyeing processes that used this machine are unavailable.

If A group of dyes belongs to a class of dyes which is not available for the case studied
Then This group of dyes is unavailable for the case studied

So, during the reasoning, the parameters of the client requirements are successively taken into account. For each criterium, all the objects that are not compatible are eliminated.

Nevertheless, even with these connections, we often obtain more than one available dyeing process, and many dyes, so that the calculation of recipes of dyes is still time consuming. The dyer often uses an other kind of knowledge.

10. The heuristical knowledge

The heuristic knowledge is the result of many years of experience, and permits to the dyer to reason more rapidly on the case to be treated. It consists of new relations between objects and new rules issued from these relations. These new relations are shown on Fig. 5.

10.1. The color

The color is a characteristic of each dye. This color depends on the dye concentration, on the fabric, and on the dyeing process. Theoretically, with the primary color (i.e., magenta, cyan and blue) plus black and

white, it is possible to obtain all the desired colors. Unfortunately, most of the time there is not such combination of dyes in groups of dyes. Nevertheless, for each group of dyes, the dyer often determines a “trichromy”. It consists of 3 or 4 dyes which permit to obtain 70–80% of the desired colors. In the 20–30% other cases, the dyer has to combine other dyes with the dyes of the trichromy. It often corresponds to color like the turkish blue, etc.

From this point, it is easy to understand that for each group of dyes, and sometimes for some classes of dyes, it is not possible to obtain all the colors. The color is then defined with terms such as dull/bright and pastel/clear/medium/dark, and the dyer use some rules such as :

If the desired color is bright
Then the class of sulphur dyes is not available

These colorimetric terms can be well-defined using the 3 dimension colorimetric space. For example, the loci of dull dyes is a cylinder with an elliptical section (Schacher, 1991).

10.2. The color fastness

This criteria are related to the chemical properties of each dyes. So, they depend on:

- the dyes, and especially the way it is fixed on the fibres,
- the fibres,
- the applications process.

Given that the color fastness is strongly related to the chemical fixation process, it is possible to define some behaviour for classes of dyes or groups of dyes. For example, most of the direct dyes cannot be used to obtain good fastness to washing at 40°C. Then, concerning this fastness, it is possible to attribute a grade to the class of direct dyes.

Nevertheless, in some cases, such as the color fastness to artificial light, this property depends on each dye. Then it is difficult to define a color fastness to a class or a group of dyes.

Whatever the fastness requirement, it has to be verified that the “match sample” fulfils it. The verification could be experimentally done by submitting the dyed sample to the fastness test. Besides, it exists algorithms that permit to evaluate the color fastness of a calculated recipe of dyes, taking into account the dyeing process, and the properties of each dye (Convert, 1997).

10.3. The dyeing properties of dyestuffs

As it is shown on Fig. 5, the link between the dyes and the article is due to the relation between the class of dyes and the fibres, and indirectly to the relation between the machine and the sub-group of articles. Using this relation, it is theoretically possible to dye all the articles that are composed of at least one of the fibres which are characteristic for the class of dyes.

For example, for the dyeing of cotton, reactive dyes are often used. They cannot be used for the dyeing of polyester fibers. Nevertheless, some of them still present a low affinity for polyester fibres, so that they can color them a little. Then, such dyes cannot be used for the dyeing of cotton/polyester fibers.

11. The meta knowledge: the general process of reasoning

The typical process of reasoning is presented on Fig. 8. The dyer always reasons in the same way. At the beginning of the reasoning, all the objects from the factory are supposed to be able to fulfil the client's requirements. The selection criteria are then ordered following their importance. Each step of the reasoning corresponds to one selection criterium, and to the elimination of the objects that cannot fulfil this criterium.

Some criteria are always considered, such as :

- the fibres that must be dyed,

- the dyeing processes that can treat the mix of fibres,
- the machine available for the article to be treated,
- the desired color,
- the quantity of article to be dyed,
- the dyes that can dye the mix of fibers,
- the reproducibility/the price.

All these criteria are considered in this order. Other criteria, that are specified in the client requirements are more or less important, so that a weight can be attributed to each of them. These criteria will then be successively considered, eventually after the three first criteria previously exposed (fibers to be dyed, process for the mix of fibers, machine available for this article).

If, after considering all the criteria, no dye or process is available, it means that the dyeing factory cannot treat the article.

At the end of this step, several dyes will be proposed to the dyer, and at least one dyeing process. Then, the possible recipe of dyes will be calculated using the color matching software.

12. Example

An example of reasoning is the following. It is carried out from client's requirements that are:

article: Polyester / Cotton 65/35 twill
 color: dark blue ($L^* = 60$, $a^* = -40$, $b^* = 12$)
 quantity to be treated: 1000 m
 washing fastness at 60°C—degradation: 4
 washing fastness at 60°C—staining: 4
 fastness to artificial light: 4

This article may have some defects of “barrel”. Moreover, this article is very fragile, so that it cannot be dyed with overflow. The sub-group of articles that it belongs to, is characterized by three dyeing machines: jigger, autoclave and pad-batch.

The dyeing factory studied often used around 230 dyestuffs. The reasoning then consists in:

- ordering the selection criteria, following their importance,
- taking successively the selection criteria into account, on a purpose to eliminate the objects that cannot fulfil these criteria.

The reasoning is given in Table 2. At the beginning, the 230 dyes are available, and the 4 dyeing machines

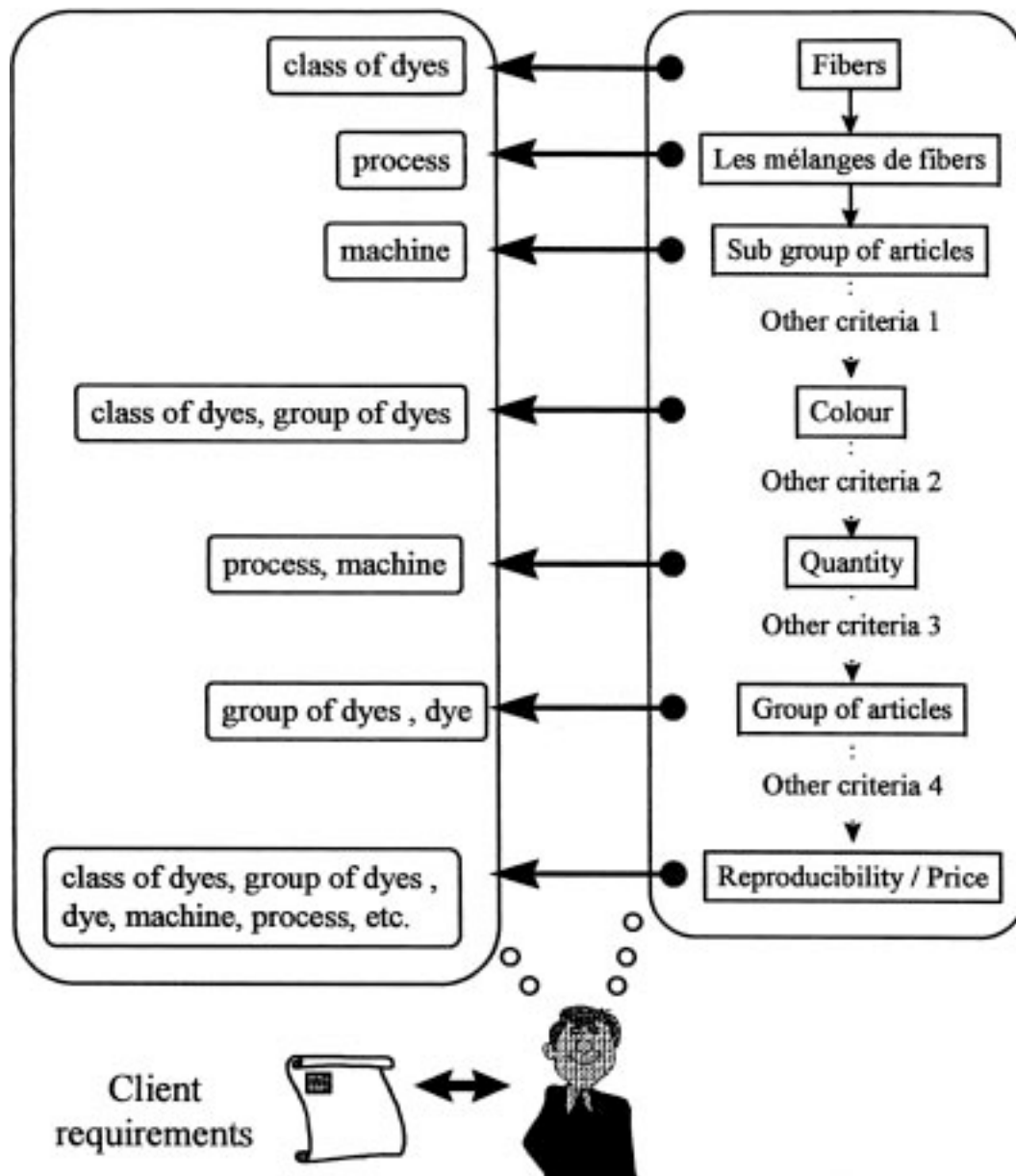


Fig. 8. The general process of reasoning.

(autoclave, jigger, overflow and pad-batch). Both cotton and polyester fibers must be dyed to obtain the same deep blue color.

At the end of the selection a recipe of dyes must be calculated for cotton fibers and polyester fibers. Most of the time, the dyes recipes consists of three dyes. Then, we obtain:

- 10 recipes of dyes possible for the 5 disperse dyes,
- 84 recipes of dyes possible for the 9 reactive dyes.

Then, the dyer selects the “best” recipe for cotton and the “best” recipe for the polyester.

Table 2. Example of reasoning

Step	Criterium	Objects available
1	Polyester/Cotton	3 classes of dyes, 9 groups of dyes, 56 dyes 16 processes, 4 machines
2	Process for the dyeing of cotton and polyester	3 classes of dyes, 9 groups of dyes, 56 dyes 9 processes, 4 machines
3	Machine available	3 classes of dyes, 8 groups of dyes, 53 dyes 4 processes, 3 machines
4	Defect possible: barres	3 classes of dyes, 4 groups of dyes, 32 dyes 4 processes, 3 machines
5	Fastness to washing at 60°C — degradation: 4	2 classes of dyes, 2 groups of dyes, 23 dyes 2 processes, 3 machines
6	Fastness to washing at 60°C — staining: 4	—————
7	Fastness to artificial light: 5	2 classes of dyes, 2 groups of dyes, 21 dyes 2 processes, 3 machines
8	Color	—————
9	Quantity	—————
10	Dyes available for mixes of cotton and polyester	2 classes of dyes, 2 groups of dyes, 14 dyes 2 processes, 3 machines
11	Reproducibility/Price	2 classes of dyes: reactive dyes for cotton and disperse dyes for polyester 2 groups of dyes: 9 reactive dyes and 5 disperse dyes

13. Conclusion

The modelization of the reasoning of an expert in dyeing is possible. The major problems are related to very subjective criteria that are considered during the work of the expert.

The model that we have proposed, developed in collaboration with a dyeing factory, uses a representation of the expertise world in a semantic net form with relations between objects established from deep knowledge but also integrating heuristic rules and factual data. It also has an exhaustive check list of all the criteria that have to be taken into account and is linked with a computer color matching system to be able to provide a high level of expertise in textile wet processing. Based on this study, it is proposed to develop interactive multimedia and hyper media training tools.

This model has been tested on several cases that were presented to the expert. Each time, the results were closed to the results of the dyer (Convert, 1998).

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